

The Unifying Theories Crib

Andrew Butterfield

February 25, 2026

Contents

1 Relations

1.1 Conditional

Definition 1.1.1 (Conditional)

$$\begin{aligned}\alpha b &\subseteq \alpha P = \alpha Q \\ P \triangleleft b \triangleright Q &=_{df} (b \wedge P) \vee (\neg b \wedge Q) \\ \alpha(P \triangleleft b \triangleright Q) &=_{df} \alpha P\end{aligned}$$

- 1.1L1 $P \triangleleft b \triangleright P = P$ (cond idemp)
- 1.1L2 $P \triangleleft b \triangleright Q = Q \triangleleft \neg b \triangleright P$ (cond symm)
- 1.1L3 $(P \triangleleft b \triangleright Q) \triangleleft c \triangleright R = P \triangleleft b \wedge c \triangleright (Q \triangleleft c \triangleright R)$ (cond assoc)
- 1.1L4 $P \triangleleft b \triangleright (Q \triangleleft c \triangleright R) = (P \triangleleft b \triangleright Q) \triangleleft c \triangleright (P \triangleleft b \triangleright R)$ (cond distr)
- 1.1L5 $P \triangleleft true \triangleright Q = P = Q \triangleleft false \triangleright P$ (cond unit)
- 1.1L6 $P \triangleleft b \triangleright (Q \triangleleft b \triangleright R) = R \triangleleft b \triangleright R$
- 1.1L7 $P \triangleleft b \triangleright (P \triangleleft c \triangleright Q) = P \triangleleft b \vee c \triangleright Q$

1.2 Composition

Definition 1.2.1 (Sequential Composition)

$$\begin{aligned}out\alpha P &= in\alpha' Q = \{v'\} \\ P(v'); Q(v) &=_{df} \exists v_0 \bullet P(v_0) \wedge Q(v) \\ in\alpha(P(v'); Q(v)) &=_{df} in\alpha P \\ out\alpha(P(v'); Q(v)) &=_{df} out\alpha Q\end{aligned}$$

- 1.2L1 $P; (Q; R) = (P; Q); R$ (; assoc)
 1.2L2 $(P \triangleleft b \triangleright Q); R = (P; R) \triangleleft b \triangleright (Q; R)$ (;-cond left distr)

1.3 Assignment

Definition 1.3.1 (Assignment)

$$\begin{aligned} \alpha e \subseteq A &= \{x, y, \dots, z, x', y', \dots, z'\} \\ x :=_A e &=_{df} (x' = e \wedge y' = y \wedge \dots \wedge z' = z) \\ \alpha(x :=_A e) &=_{df} A \end{aligned}$$

- 1.3L1 $(x := e) = (x, y := e, y)$
 1.3L2 $(x, y, z := e, f, g) = (y, x, z := f, e, g)$
 1.3L3 $(x := e; x := f(x)) = (x := f(e))$
 1.3L4 $x := e; (P \triangleleft b(x) \triangleright Q) = (x := e; P) \triangleleft b(e) \triangleright (x := e; Q)$

Definition 1.3.2 (Skip)

$$\begin{aligned} A &= \{v, v'\} \\ \mathbb{I}_A &=_{df} (v' = v) \\ \alpha \mathbb{I}_A &=_{df} A \end{aligned}$$

- 1.3L5 $P; \mathbb{I}_{\alpha P} = P = \mathbb{I}_{\alpha P}; P$ (; unit)

Definition 1.3.3 (Array Assignment)

$$w[i] := e =_{df} w \oplus \{i \mapsto e\}$$

1.4 Non-determinism

Definition 1.4.1 (Non-deterministic choice)

$$\begin{aligned} \alpha P &= \alpha Q \\ P \sqcap Q &=_{df} P \vee Q \\ \alpha(P \sqcap Q) &=_{df} \alpha P \end{aligned}$$

1.4L1	$P \sqcap Q = Q \sqcap P$	($\sqcap$ symm)
1.4L2	$P \sqcap (Q \sqcap R) = (P \sqcap Q) \sqcap R$	($\sqcap$ assoc)
1.4L3	$P \sqcap P = P$	($\sqcap$ idemp)
1.4L4	$P \sqcap (Q \sqcap R) = (P \sqcap Q) \sqcap (P \sqcap R)$	($\sqcap$ distr)
1.4L5	$P \triangleleft b \triangleright (Q \sqcap R) = (P \triangleleft b \triangleright Q) \sqcap (P \triangleleft b \triangleright R)$	(cond- $\sqcap$ distr)
1.4L6	$(P \sqcap Q); R = (P; R) \sqcap (Q; R)$	(;- $\sqcap$ left distr)
1.4L7	$P; (Q \sqcap R) = (P; Q) \sqcap (P; R)$	(;- $\sqcap$ right distr)
1.4L8	$P \sqcap (Q \triangleleft b \triangleright R) = (P \sqcap Q) \triangleleft b \triangleright (P \sqcap R)$	($\sqcap$ -cond distr)

Definition 1.4.2 (Abort)

$$\begin{aligned}\perp_A &=_{df} \mathbf{true} \\ \alpha \perp_A &=_{df} A\end{aligned}$$

1.5 The complete lattice of relations

- 1.5L1 $[P \Leftarrow \sqcap S]$ **iff** $([P \Leftarrow X] \text{ for all } X \text{ in } S)$
1.5L1A $[\sqcap S \Leftarrow X]$ for all X in S
1.5L1B If $[P \Leftarrow X]$ for all X in S , then $[P \Leftarrow \sqcap S]$

$$\sqcap \{\} = \mathbf{false}$$

Definition 1.5.1 (Miracle)

$$\begin{aligned}\top_A &=_{df} \mathbf{false} \\ \alpha \top_A &=_{df} A\end{aligned}$$

$$[P \Rightarrow \sqcup S] \text{ **iff** } ([P \Rightarrow X] \text{ for all } X \text{ in } S)$$

$$\sqcup \{\} = \mathbf{true}$$

- 1.5L2 $(\sqcup S) \sqcap Q = \sqcup \{ P \sqcap Q \mid P \in S \}$
1.5L3 $(\sqcap S) \sqcup Q = \sqcap \{ P \sqcup Q \mid P \in S \}$
1.5L4 $(\sqcap S); Q = \sqcap \{ P; Q \mid P \in S \}$ (; left univ disj)
1.5L5 $R; (\sqcap S) = \sqcap \{ R; P \mid P \in S \}$ (; right univ disj)

1.6 Recursion

Definition 1.6.1 (Weakest fixed point)

$$\mu F =_{df} \sqcap \{ X \mid [X \Rightarrow F(X)] \}$$

- 1.6L1 $[Y \Rightarrow \mu F]$ whenever $[Y \Rightarrow F(Y)]$ (weakest fixed point)
 1.6L2 $[F(\mu F) \equiv \mu F]$ (fixed point)

Definition 1.6.2

$$\begin{aligned} \mu X \bullet \dots X \dots X \dots &=_{df} \mu F \\ \text{where } F &=_{df} \lambda X \dots X \dots X \dots \\ b * P &=_{df} \mu X \bullet ((P; X) \triangleleft b \triangleright I) \end{aligned}$$

Exercise 1.6.4

$$\begin{aligned} \mu X \bullet F(G(X)) &= F(\mu X \bullet G(F(X))) \\ \text{If } [G(H(X)) \Rightarrow F(G(X))] &\text{ for all } X, \text{ then } [G(\mu H) \Rightarrow F] \end{aligned}$$

1.7 Strongest fixed point

Definition 1.7.1 (Strongest fixed point)

$$\nu F =_{df} \neg \mu X \bullet \neg F(\neg X)$$

- 1.7L1 $[\nu F \Rightarrow S]$ whenever $[F(S) \Rightarrow S]$
 1.7L2 $[F(\nu F) \equiv F]$
 1.7L3 $\text{false}; P = \text{false} = P; \text{false}$

Lemma 1.7.2

$$\begin{aligned} \text{If } [F(C \Rightarrow S) \Rightarrow (C \Rightarrow S)] &\text{ and } [C \Rightarrow (\mu F \equiv \nu F)] \\ \text{then } [\mu F \Rightarrow (C \Rightarrow S)] \end{aligned}$$

Definition 1.7.3 (Approximation chain)

A set of predicates $E = \{E_i \mid i \in \mathcal{N}\}$ is called an approximation chain for C if

$$\begin{aligned} E_0 &= \text{false} \\ [E_i \Rightarrow E_{i+1}] &\text{, for all } i \text{ in } \mathcal{N} \\ C &= \bigvee_i E_i \end{aligned}$$

Lemma 1.7.4

$$\begin{aligned} \text{If } X \wedge E_n &= Y \wedge E_n, \text{ for all } n \text{ in } \mathcal{N} \\ \text{then } X \wedge (\bigvee_n E_n) &= Y \wedge (\bigvee_n E_n) \end{aligned}$$

Definition 1.7.5 (E -constructive)

Let E be an approximation chain of C . A function F is E -constructive if. for all X and n :

$$F(X) \wedge E_{n+1} = F(X \wedge E_n) \wedge E_{n+1}$$

For such E and F :

$$C \wedge \mu F = C \wedge \nu F$$

1.8 Preconditions and postconditions

Definition 1.8.1 (Hoare triple)

$$p\{Q\}r \quad =_{df} \quad [Q \Rightarrow (p \Rightarrow r')]$$

- 1.8L1 If $p\{Q\}r$ and $p\{Q\}s$ then $p\{Q\}(r \wedge s)$
- 1.8L2 If $p\{Q\}r$ and $q\{Q\}r$ then $(p \vee q)\{Q\}r$
- 1.8L3 If $p\{Q\}r$ then $(p \wedge q\{Q\})(r \vee s)$
- 1.8L4 $r(e)\{x := e\}r(x)$
- 1.8L5 If $(p \wedge b)\{Q1\}r$ and $(p \wedge \neg b)\{Q2\}r$ then $p\{Q1 \triangleleft b \triangleright Q2\}r$
- 1.8L6 If $p\{Q1\}s$ and $s\{Q2\}r$ then $p\{Q1; Q2\}r$
- 1.8L7 If $p\{Q1\}r$ and $p\{Q2\}r$ then $p\{Q1 \sqcap Q2\}r$
- 1.8L8 If $b \wedge c\{Q\}c$ then $c\{\nu X \bullet Q; X \triangleleft b \triangleright \mathbb{I}\}(\neg b \wedge c)$
- 1.8L9 $false\{Q\}r$ and $p\{Q\}true$ and $p\{false\}false$ and $p\{\mathbb{I}\}p$

Definition 1.8.3 (Floyd assertion and assumption)

$$\begin{aligned} c_{\perp} &=_{df} \quad \mathbb{I} \triangleleft c \triangleright \perp && \text{(assertion)} \\ c^{\top} &=_{df} \quad \mathbb{I} \triangleleft c \triangleright \top && \text{(assumption)} \end{aligned}$$

Definition 1.8.4 (Weakest precondition)

$$Q \text{ wp } r \quad =_{df} \quad \neg (Q; \neg r)$$

- 1.8L10 $(x := e) \text{ wp } r(x) = r(e)$
- 1.8L11 $(P; Q) \text{ wp } r = P \text{ wp } (Q \text{ wp } r)$
- 1.8L12 $(P \triangleleft b \triangleright Q) \text{ wp } r = (P \text{ wp } r) \triangleleft b \triangleright (Q \text{ wp } r)$
- 1.8L13 $(P \sqcap Q) \text{ wp } r = (P \text{ wp } r) \wedge (Q \text{ wp } r)$
- 1.8L14 If $[r \Rightarrow s]$ then $[Q \text{ wp } r \Rightarrow Q \text{ wp } s]$
- 1.8L15 If $[Q \Rightarrow S]$ then $[S \text{ wp } r \Rightarrow Q \text{ wp } r]$ (antimontonicity)
- 1.8L16 $Q \text{ wp } (\bigwedge R) = \bigvee \{ Q \text{ wp } r \mid r \in R \}$
- 1.8L17 $Q \text{ wp } false = false$ provided $Q; true = true$
- 1.8L18 $[(P; Q) \Rightarrow S] \text{ iff } [P \Rightarrow Q \text{ wp } S]$

Exercise 1.8.5

$$\begin{aligned} b_{\perp}; c_{\perp} &= (b \wedge c)_{\perp} = b_{\perp} \sqcap c_{\perp} \\ b^{\top}; c^{\top} &= (b \wedge c)^{\top} = b^{\top} \sqcup c^{\top} \\ b^{\top}; b_{\perp} &= b^{\top} \end{aligned}$$

1.9 Variable declarations

Definition 1.9.1 (Declaration and undeclaration)

$$\begin{aligned} x, x' &\in A \\ \mathbf{var} \ x &=_{df} \exists x \bullet \mathbb{I}_A \\ \mathbf{end} \ x &=_{df} \exists x' \bullet \mathbb{I}_A \\ \alpha(\mathbf{var} \ x) &=_{df} A \setminus \{x\} \\ \alpha(\mathbf{end} \ x) &=_{df} A \setminus \{x'\} \end{aligned}$$

$$\begin{aligned} &\mathbf{var} \ x; Q = \exists x \bullet Q \\ &Q; \mathbf{end} \ x; Q = \exists x' \bullet Q \\ 1.9L1 \quad &(\mathbf{var} \ x; \mathbf{var} \ y) = (\mathbf{var} \ y; \mathbf{var} \ x) = \mathbf{var} \ x, y \\ 1.9L2 \quad &(\mathbf{end} \ x; \mathbf{end} \ y) = (\mathbf{end} \ y; \mathbf{end} \ x) = \mathbf{end} \ x, y \\ 1.9L3 \quad &(\mathbf{var} \ x; \mathbf{end} \ y) = (\mathbf{end} \ y; \mathbf{var} \ x) \text{ provided } x, y \text{ distinct} \\ 1.9L4 \quad &\text{If } x : T \text{ then } \mathbf{var} \ x = \sqcap \{ \mathbf{var} \ x := k \mid k \in T \} \\ 1.9L5 \quad &\mathbf{var} \ x; \mathbf{end} \ x = \mathbb{I} \\ 1.9L6 \quad &\mathbf{var} \ x; \mathbf{end} \ x = \mathbb{I} \\ 1.9L7 \quad &(\mathbf{end} \ x; \mathbf{var} \ x := e) = (x := e) \text{ provided } x \text{ not in } e \\ 1.9L8 \quad &(x := e; \mathbf{end} \ x) = \mathbf{end} \ x \end{aligned}$$

Definition 1.9.5 (Alphabet extension)

$$\begin{aligned} x, x' &\notin \alpha R \\ R_{+x} &=_{df} R \wedge (x' = x) \\ \alpha R_{+x} &=_{df} \alpha R + \{x, x'\} \end{aligned}$$

If R does not mention x :

$$\begin{aligned} 1.9L9 \quad &\mathbf{var} \ x; R_{+x}; P; \mathbf{end} \ x = R; \mathbf{var} \ x; P; \mathbf{end} \ x \\ 1.9L2 \quad &\mathbf{var} \ x; P; R_{+x}; \mathbf{end} \ x = \mathbf{var} \ x; P; \mathbf{end} \ x; R \end{aligned}$$

Exercise 1.9.6

$$\begin{aligned} (\mathbf{var} \ x) \mathbf{wp} \ r &= \forall x \bullet r \\ (\mathbf{end} \ x) \mathbf{wp} \ r &= r, \quad \text{if } x \text{ is not free in } r \end{aligned}$$